EFS Learning Academy — Interactive 3D Equipment & Maintenance Training

Proposal for the Centre of Excellence (CoE) | Live prototype: browser-based, hosted on Render, source on GitHub | October 2026

I have designed and built a browser-based training platform in which technicians explore equipment as interactive 3D models and practise maintenance tasks step by step — first watching a guided demonstration, then performing the task themselves using a mouse, touch, or their own hands via the webcam. It runs on a standard laptop or tablet with no installation, plug-ins or paid licences.

1. Technology Stack (as built)

Layer	Technology	Purpose / design decision
Front end	JavaScript (ES modules), HTML5, CSS3	Framework-free for a small, fast bundle; responsive layout for desktop, tablet and mobile
3D engine	Three.js r186 (WebGL)	Procedurally modelled machines, exploded views, cut-aways, airflow and water effects, smooth camera framing
Hand tracking	Google MediaPipe Tasks Vision (WebAssembly, Web Worker)	First-person hand control (grab, carry, press). Processed fully on-device — no video leaves the browser
Voice guidance	Web Speech API	Machine Guide & Page Guide narration in English, Hinglish and Hindi, sentence-complete playback
Build & tooling	Node.js 22, esbuild	Code-splitting, content-hashed files, Brotli/gzip pre-compression, lazy-loaded 3D and camera modules
Quality assurance	Node test runner, Playwright + Chromium	102 automated unit tests; 43 end-to-end browser checks covering every 3D task, demo and practice
Hosting & CI/CD	GitHub, Render (static site)	Auto-deploy from *main*; security headers (CSP, Permissions-Policy, nosniff); no external network calls
Engineering assistant	Claude Code (AI coding agent)	Implementation, code review, performance tuning and automated testing under my direction

Delivered: 9 equipment modules (Soft Services: Junior 103, Junior 429, HYPER 2000B; Hard Services: 1-/3-phase motors, alternator, megger, multimeter, HVAC) and a fully 3D Maintenance Checklist — filter wash (dry/wet), dust-bag replacement and connector inspection with safe-response decision.

2. Application at the Centre of Excellence — Implementation Plan

Phase	Timeline	Key activities	Outcome / KPI
1. Pilot	Month 1	Deploy to CoE trainers; run with one induction batch; SME & HSE review of content against manufacturer manuals	Content sign-off; trainer & learner feedback score
2. Content scale-up	Months 2–3	Add highest-volume equipment (single-disc, pressure washer, HVAC PPM, DB visual inspection) using the reusable task framework	10+ machines; pre-site readiness check in place
3. Platform	Months 4–6	Backend, RBAC, SSO, progress tracking and assessments (see section 4)	Tracked completion rate, assessment pass rate
4. Roll-out	Month 6+	Make completion a prerequisite before handling equipment on site; extend to regional training hubs	Reduced equipment misuse incidents & induction time

CoE use cases: new-hire induction • refresher before site deployment • toolbox talks • pre-assessment before practical sign-off • multilingual workforce training without needing a physical machine or downtime.

3. Requirements to Progress

Functional & non-functional requirements	Resources and tools required
• Content validated by Subject Matter Experts and HSE before release • Runs on Chrome/Edge on standard company laptops and tablets • Camera processing on-device only; compliant with company data-privacy policy • Accessibility to WCAG 2.1 AA; keyboard, mouse, touch and hand input • Clear disclaimer: training aid, not a certification or service manual	• SME / trainer time (approx. 2–4 hrs per new machine) and manufacturer manuals • Reference photos and dimensions of equipment in the CoE workshop • Test devices: 2 webcam laptops, 1 tablet; company-approved cloud hosting • Company GitHub organisation repository and Claude Code licence • IT support for Microsoft Entra ID (SSO) and a managed PostgreSQL database

4. Future Scope — Backend & Role-Based Access Control (RBAC)

Backend architecture

- **API:** Node.js (NestJS) REST services, containerised (Docker)
- **Database:** PostgreSQL — users, sites, courses, attempts, scores, audit log
- **Auth:** Microsoft Entra ID SSO (OAuth 2.0 / OIDC), JWT sessions
- **Reporting:** dashboards by site, role and machine; CSV / Power BI export
- **Integration:** xAPI / SCORM for the corporate LMS; content admin panel

Role	Access
Administrator	Users, roles, sites, system settings, audit log
CoE Trainer	Create/edit modules, assign training, review results
Site Supervisor	Track and assign training for own team only
Technician	Complete assigned modules and assessments
Auditor / HSE	Read-only access to records and compliance reports

Further enhancements: scored assessments with completion certificates • offline mode (PWA) for sites with limited connectivity • Arabic language support • higher-fidelity models from manufacturer CAD • VR/AR headset support (WebXR).